

PROFORMA — Internal Enterprise Tools Infrastructure Billing

LIONGATE SARL — Infrastructure Cost Estimation

Document: PROFORMA-04

Context: Internal Tools (HR, CRM-lite, dashboards, ops portals, employee apps)

Hosting Provider: Hetzner Cloud

Currency: EUR (€)

Version: 1.0

How to read this document:

This proforma covers infrastructure to host **internal tools used by LIONGATE staff only** — typically behind VPN/SSO, low concurrency (5–80 users), but business-critical.
Three tiers are presented because high-availability is rarely required for internal tools.

Reference Stack

Layer	Technology	Notes
Frontend	Next.js / Refine / React Admin	Admin-style UIs
Backend	NestJS / Django / FastAPI	REST or RPC
Database	PostgreSQL 16	Shared cluster for multiple tools
Auth	Authentik / Keycloak / Cloudflare Access	SSO for all internal apps
VPN	WireGuard / Tailscale (free for ≤ 3 users)	Private access
Reverse proxy	Nginx / Traefik	Routes per-subdomain
Container	Docker Compose	One node hosts multiple tools
Storage	Hetzner Volume + Storage Box (backups)	Shared

Layer	Technology	Notes
Monitoring	Uptime Kuma (self-hosted)	Free, lightweight

 Internal tools are normally **co-located on a shared internal-apps server** to minimise cost.
Each tool runs as its own Docker stack with its own database schema.

Tier 1 – Minimum Viable Internal Stack (Single shared host)

One internal-apps server hosting up to ~5 small tools + shared Postgres + shared auth.
Audience: ≤ 15 employees. **Concurrency:** ≤ 10 simultaneous users.

Production (single shared host)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	Internal Apps Server	CX32 — 4 vCPU, 8 GB RAM, 80 GB NVMe	€8.21	1	8.21	98.52
2	Hetzner Volume (shared data)	100 GB	€5.20	1	5.20	62.40
3	Storage Box (backups)	BX11 — 100 GB	€3.43	1	3.43	41.16
4	Floating IP	Static IPv4	€2.00	1	2.00	24.00
5	SSL	Let's Encrypt wildcard	Free	1	0.00	0.00
6	Internal subdomain (*.tools.liongate.app)	Same domain zone	Included	1	0.00	0.00
7	SSO / Auth (Authentik self-hosted)	Runs on same server	Free	1	0.00	0.00

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
8	VPN (Tailscale Free ≤ 3 users)	Or WireGuard self-hosted	Free	1	0.00	0.00
9	Outbound email	Brevo Free — 300/day	Free	1	0.00	0.00
10	Uptime Kuma (self-hosted)	Same server	Free	1	0.00	0.00
—	SUBTOTAL				€18.84	€226.08

Tier 1 — Total Summary

Environment	Monthly (€)	Annual (€)
Production	18.84	226.08
Staging	<i>(not used — preview locally or on dev branch deploys)</i>	—
Development	0.00 (local Docker)	0.00
TOTAL TIER 1	€18.84	€226.08

Tier 2 — Recommended Internal Stack (Split DB + SSO)

App server hosts internal tools; PostgreSQL on its own server; Authentik in its own container set.

Audience: 15–60 employees. **Concurrency:** 10–30 simultaneous users.

Production (Recommended)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	Internal Apps Server	CX42 — 8 vCPU, 16 GB RAM	€17.77	1	17.77	213.24

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
2	DB Server (shared across tools)	CX32 — 4 vCPU, 8 GB RAM	€8.21	1	8.21	98.52
3	Volume — DB data	100 GB	€5.20	1	5.20	62.40
4	Volume — Apps data / file uploads	100 GB	€5.20	1	5.20	62.40
5	Private Network (VLAN)	App ↔ DB	Free	1	0.00	0.00
6	Storage Box (backups)	BX21 — 500 GB	€5.94	1	5.94	71.28
7	Floating IP	Static IPv4	€2.00	1	2.00	24.00
8	SSL	Let's Encrypt wildcard	Free	1	0.00	0.00
9	SSO (Authentik self-hosted on app server)	Free	Free	1	0.00	0.00
10	Outbound email	Brevo Starter — 20k/mo	€19.00	1	19.00	228.00
11	VPN (Tailscale Premium, 10 users)	Per user	€6.00	10	60.00	720.00
12	Snapshots (weekly ×2)	~40 GB	€0.48	1	0.48	5.76
—	SUBTOTAL				€123.80	€1,485.60

 If using Tailscale Free (max 3 users) or self-hosted WireGuard, **save €60/mo**.
Recommended saving: self-host WireGuard → reduces Tier 2 to **€63.80/mo (€765.60/yr)**.

Tier 2 — Total Summary

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Production	63.80 <i>(self-hosted VPN)</i>	123.80	765.60	1,485.60

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Staging	<i>(branch deploys on app srv)</i>	—	—	—
Development	0.00 (local Docker)	0.00	0.00	0.00
TOTAL TIER 2	€63.80	€123.80	€765.60	€1,485.60

Tier 3 — Full Internal Stack (Production-grade)

Dedicated SSO server, dedicated monitoring, dedicated backup orchestration, dedicated CI runner.
Audience: 60+ employees, multiple departments, business-critical tools (HR, finance).

Production (Full)

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
1	Internal Apps Server	CX42 — 8 vCPU, 16 GB RAM	€17.77	1	17.77	213.24
2	DB Server	CX42 — 8 vCPU, 16 GB RAM	€17.77	1	17.77	213.24
3	SSO Server (Authentik / Keycloak)	CX22	€4.35	1	4.35	52.20
4	Monitoring Server (Grafana + Uptime)	CX22	€4.35	1	4.35	52.20
5	Self-hosted CI runner	CX32	€8.21	1	8.21	98.52
6	Volume — DB data	200 GB	€10.40	1	10.40	124.80

#	Resource	Spec	Unit Price	Qty	Monthly (€)	Annual (€)
7	Volume — Apps shared data	200 GB	€10.40	1	10.40	124.80
8	Storage Box (backups)	BX31 — 1 TB	€8.69	1	8.69	104.28
9	Floating IP × 2	Static IPv4 (apps + SSO)	€2.00	2	4.00	48.00
10	Private Network	VLAN	Free	1	0.00	0.00
11	SSL	Let's Encrypt wildcard	Free	1	0.00	0.00
12	Outbound email	Brevo Business — 100k/mo	€49.00	1	49.00	588.00
13	Tailscale Business (per user)	25 users	€18.00	25	450.00	5,400.00
14	External uptime monitoring	UptimeRobot Pro	€7.00	1	7.00	84.00
15	Snapshots (rotation ×4)	~80 GB	€0.95	1	0.95	11.40
—	SUBTOTAL				€596.89	€7,114.68

💡 **Tailscale is the biggest single line item.** Alternative recommended for cost-sensitive companies:

- Self-host WireGuard or Headscale → near-zero ongoing cost
- Use Cloudflare Access (Zero Trust Free up to 50 users) → €0 instead of €450/mo

Replacing Tailscale Business with **Cloudflare Access Free** reduces Tier 3 to **€146.89/mo (€1,762.68/yr)**.

Tier 3 — Total Summary

Environment	Monthly MIN (€)	Monthly MAX (€)	Annual MIN (€)	Annual MAX (€)
Production	146.89 (CF Access Free)	596.89	1,762.68	7,114.68
Staging	(branch deploys)	8.21	—	98.52
Development	0.00 (local Docker)	0.00	0.00	0.00
TOTAL TIER 3	€146.89	€605.10	€1,762.68	€7,213.20

Internal Tools — Tier Comparison Dashboard

Tier	Setup Type	Monthly MIN	Monthly MAX	Annual MIN	Annual MAX	Best for
T1	Single shared host	€18.84	€18.84	€226	€226	≤ 15 employees, < 5 tools
T2	Split DB + SSO + VPN	€63.80	€123.80	€766	€1,486	15–60 employees, business tools
T3	Full stack with SSO, CI, monitoring	€146.89	€605.10	€1,763	€7,213	60+ employees, critical workflows

Tool Co-location Strategy (Cost Optimisation)

Internal tools should **share infrastructure** wherever possible. Recommended co-location pattern:

Server	Hosts
internal-apps-01	All internal Next.js/NestJS apps via Docker Compose + Traefik
internal-db-01	Single Postgres cluster with one schema per tool
sso-01	Authentik / Keycloak (Tier 3 only)
ops-01	Grafana, Loki, Prometheus, Uptime Kuma
ci-01	Self-hosted GitHub Actions runner

 Adding one more internal tool to T2 costs **€0 of extra infrastructure** if app server has spare RAM.
Only DB size and backup volume grow. **Real marginal cost: ~€2–€5/mo per added tool.**

Hidden & Variable Costs

Cost Item	Frequency	Estimated Amount	Notes
Marginal new internal tool	Monthly	+€2–€5	If co-located on shared host
VPN seat growth	Monthly	+€6–€18 per user	Tailscale paid tiers
DB volume growth	Monthly	+€0.52 per +10 GB	Hetzner volume
SSO licence (Authentik Enterprise)	Annual	€0 (community) / €5,000+ (enterprise)	Community is sufficient for ≤ 100 users
Storage Box upgrade	Monthly	+€2–€6	When backups grow past 1 TB
Audit log retention	Monthly	+€2–€10	Loki/MinIO storage

Budget Planning Summary (Internal Tools)

LIONGATE SARL – Internal Tools Budget Reference

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MINIMUM MONTHLY BUDGET: €18.84 (Tier 1 – single shared host)
RECOMMENDED MONTHLY BUDGET: €63.80 (Tier 2 – self-hosted VPN)
FULL PRODUCTION BUDGET: €146.89 (Tier 3 – Cloudflare Access Free)
PREMIUM BUDGET (Tailscale): €596.89 (Tier 3 – full Tailscale Business)

MINIMUM ANNUAL BUDGET: €226.08
RECOMMENDED ANNUAL BUDGET: €765.60
FULL PRODUCTION ANNUAL: €1,762.68
PREMIUM ANNUAL: €7,114.68

CONTINGENCY BUFFER (+10%, low volatility):
Tier 2 recommended: €765.60 × 1.10 = €842.16/yr
Tier 3 full: €1,762.68 × 1.10 = €1,938.95/yr

Decision Guide – Which tier for internal tools?

Question	If YES → recommend
Fewer than 15 staff, 1–4 small internal tools	Tier 1
HR/Finance/CRM-lite tools used daily by ≥ 15 staff	Tier 2
Multi-department, role-based SSO required	Tier 3
Need GitHub Actions self-hosted runner	Tier 3
Compliance/audit logging required	Tier 3 + Loki retention
Want zero ongoing VPN cost	Use Cloudflare Access Free

Review: Annually or when staff count crosses tier threshold

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